Sahaj Bhadja

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Summary

Full Stack Software Engineer with 2+ years of experience building **React applications**, **Python/FastAPI services**, PostgreSQL-backed workflows, internal tools, and reliable customer-facing financial products.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Reduced dashboard latency from 1.2s to 350ms through **SQL indexing and query optimization**, designing normalized schemas and row-level isolation for a multi-tenant education platform supporting 500+ users reliably.
- Tripled concurrent throughput across 1K+ AI jobs by restructuring **FastAPI background workflows** with explicit state transitions, failure isolation, recoverable processing, and observable services under sustained production load.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Processed 1K+ files per batch using **Python multithreading**, validating heterogeneous inputs while isolating failures and preserving responsive customer workflows during concurrent model execution and production traffic reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while troubleshooting integrations across Windows, Linux, and macOS production environments reliably at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins and GitHub**, adding failure reports and cross-platform validation across Linux, Windows, and macOS with repeatable nightly workflows reliably daily.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Synchronized 2M+ records through an eight-service **React and Node.js platform** covering authentication, resources, customers, billing, leads, permissions, and notifications with WebSockets and Azure SQL across locations and workflows.
- Reduced resource wastage by 42% through **real-time data synchronization**, implementing role-based access, WebSocket updates, automated tests, Azure monitoring, and Docker/Jenkins deployments across eight modular product service domains.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Delivered **sub-200ms Flask APIs** across 50K+ booking records, implementing pricing, coupon validation, payment webhooks, and Firebase status synchronization while using transactions and locking to prevent conflicting reservations.
- Reduced manual refund handling by approximately 70% through **payment workflow automation**, supporting refunds, disputes, and consistent status while presenting architecture and backend integration workflows directly to clients reliably.

Skills

Languages: Python, TypeScript, JavaScript, SQL, Java, HTML, CSS
Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs, Microservices, API Design, WebSockets
Frontend: React, React Native, Next.js, Material UI, Tailwind CSS, Responsive Design, Reusable Components
Data & Databases: PostgreSQL, MySQL, MongoDB, Redis, AWS RDS, Azure SQL, Firebase, Data Modeling, Query Optimization
Cloud & Delivery: AWS, GCP, Azure, Docker, Kubernetes, Jenkins, CI/CD, Git, GitHub
Quality & Design: Data Structures, Algorithms, Object-Oriented Design, System Design, Unit Testing, Integration Testing, PyTest, Production Debugging

Projects

Insurance Program Operations Platform - Design Concept

- Designed a **React and FastAPI insurance platform** concept with configurable program workflows, broker submissions, policy status, PostgreSQL persistence, role-based access, validation, audit history, and automated integration tests.

Claims and Risk Decision Dashboard - Design Concept

- Designed a **Python risk-data workflow** concept combining validated application inputs, decision rules, claims indicators, explainable outputs, review queues, reusable React components, and operational monitoring for consistent decisions.

FinNexus | Live

- Built a **secure fintech platform** supporting 50K+ users and 1M+ transactions with Plaid bank linking, token-based access controls, real-time synchronization, advanced filtering, Kubernetes deployment, and MongoDB-backed data access.